INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Infosys SE Complete Preparation — Clocks and Calendar Logic

Section: prepinsta infosys SE | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Infosys SE Complete Preparation — Clocks and Calendar Logic

Source Portal: PrepInsta Infosys Preparation Suite (`https://prepinsta.com/infosys/logical/clocks-and-calendar/`)
Target Role: Infosys Systems Engineer (SE) | **Category:** 02_Logical_Reasoning_Ability
Total Verified Questions Extracted: 10
Quizzes Covered: quiz-1, quiz-2, quiz-3, quiz-4

📋 Section 1: Rules to Keep in Mind

These rules and fundamental principles govern all Infosys questions on this topic:

- Clock Angle Formula: Angle theta = |30H - (11/2)M|.
- Minute hand speed = 6 degrees/min; Hour hand speed = 0.5 degrees/min; Relative speed = 5.5 degrees/min.
- Coincidence (0 degrees): Occurs 11 times in 12 hours, 22 times in 24 hours.
- Opposite (180 degrees): Occurs 11 times in 12 hours, 22 times in 24 hours.
- Right angles (90 degrees): Occurs 22 times in 12 hours, 44 times in 24 hours.
- Calendar Odd Days: Ordinary year = 1 odd day (365 = 52 weeks + 1 day); Leap year = 2 odd days.
- 100 years = 5 odd days; 200 years = 3 odd days; 300 years = 1 odd day; 400 years = 0 odd days.

⚡ Section 2: How to Solve Quickly

Proven time-saving strategies recommended by PrepInsta for Infosys 2026 examination:

1. **Option Elimination via Unit Digits & Multiples:** Before doing full calculations, inspect the options to see if last digits or divisibility by 3, 7, 9, 11 can eliminate 2 to 3 wrong choices.
2. **Approximation & Magnitude Checks:** When exact arithmetic is tedious, round numbers to neat multiples (e.g., 99 ≈ 100) to find the nearest valid option.
3. **Reverse Option Substitution:** In complex algebraic equations, plug option values directly into the question stem to test validity.

🚀 Section 3: Tricks & Shortcuts

Key mathematical formulas, mental-math shortcuts, and speed axioms:

- Day of the week codes: 0=Sunday, 1=Monday, 2=Tuesday, 3=Wednesday, 4=Thursday, 5=Friday, 6=Saturday.
- A century cannot end on Tuesday, Thursday, or Saturday.

📑 Section 4: Practice Questions, Options & Step-by-Step Solutions

Contains **10** verbatim questions with options, verified answer keys, and complete derivations:

Question 1 (PrepInsta quiz-1 - Q1)

Question Stem:
In a particular year, the month of january had exactly 4 thursdays, and 4 sundays. On which day of the week did january 1st occur in the year?

Options:
- **(A)** `Monday`
- **(B)** `Tuesday`
- **(C)** `Wednesday`
- **(D)** `Thursday`

Correct Answer: (A) `Monday`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
If a month has 31 days, and it starts with sunday, Then Sundays,

Mondays, tuesdays are 5 for that month. If this month starts

with monday, then mondays, tuesdays, and wednesdays

are 5 and remaining days are 4 each. so this month start with Monday.
```

Question 2 (PrepInsta quiz-1 - Q2)

Question Stem:
What is the angle between two hands when time is 5:30?

Options:
- **(A)** `20 Degrees`
- **(B)** `18.75 Degrees`
- **(C)** `160 Degrees`
- **(D)** `15 Degrees`

Correct Answer: (D) `15 Degrees`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
 


At 5:30 minute hand will be at position 6 and hour hand will be exactly between


5 and 6 and will cover 15°, so angle between two hands=30 -15 = 15°
```

Question 3 (PrepInsta quiz-1 - Q3)

Question Stem:

The famous church in the city of Kumbakonam has a big clock tower and is said to be over 300 years old. Every Monday 10.00 A M the clock is set by Antony, doing service in the church. The Clock loses 6 minutes every hour. What will be the actual time when the faulty clock shows 3 P.M on Friday?

Options:

- **(A)** `1.06 am.`
- **(B)** `1.40 am.`
- **(C)** `12.10 am.`
- **(D)** `12.10 pm.`

Correct Answer: (A) `1.06 am.`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let us start from Monday 10 am to friday 3.00 pm.....total hours=


4 days 5 hrs.=4*24+5=101 hrs.


so total time loses=101*6 minutes=606 minutes=606/60=10 hrs 6 minutes


so actual time=friday 3.00 pm +10 hrs 06 minutes=staurday 1.06 am.


Now, in most of the places they are giving answers as 4.54 AM, which is incorrect. As the clock
loses 6 mins. Thus we have to add the time.
```

Question 4 (PrepInsta quiz-1 - Q4)

Question Stem:

How many times do the hands of a clock coincide in 5 hours?

Options:

- **(A)** `5`
- **(B)** `4`

- **(C)** `6`
- **(D)** `3`

Correct Answer: (B) `4`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
hour hand and minute hand coincide every 65:5 minute, so in 5 hour they will


coincide 4 times


The hour hand covers 2*360 degrees in 24 hours, i.e 1440 minutes.


So, in one minute, it covers 1440/720 = 1/2 degree.


Difference in angular distance travelled by the minute hand


and the hour hand in one minute is thus 6-1/2 = 11/2 degrees.


So, on a full rotation (360°), any similar event between them


will be repeated every 360/(11/2) = 65 5/11 minutes.
```

Question 5 (PrepInsta quiz-1 - Q5)

Question Stem:

What is the chance that a leap year selected at random contains 53 Fridays?

Options:

- **(A)** `1/7`
- **(B)** `3/7`
- **(C)** `2/7`
- **(D)** `2/13`

Correct Answer: (C) `2/7`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
ans- 2/7


in aleap year there are 366 days means 52 weeks and 2days. so already we have


52 fridays. now the rest two days can be


(sun, mon), (mon,tues),(tues, wed), (wed, thrus), (thrus, fri),


(fri, sat), (sat, sun)


so, the probability of 53 fridays= 2/7
```

Question 6 (PrepInsta quiz-1 - Q6)

Question Stem:

Between 9 a.m and 9 p.m of a particular day for how many times are the minute and hour hands together?

Options:

- **(A)** `11`
- **(B)** `12`
- **(C)** `13`
- **(D)** `9`

Correct Answer: (A) `11`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
&amp;nbsp;


11..the hands will be together for once per hour..e.g 9.45,10.50..but between 11 and 12 the coming together will be counted as 1 for both hence in 12 hrs time they will be together for 12-1=11 times
```

Question 7 (PrepInsta quiz-1 - Q7)

Question Stem:

How many palindromes are there in a clock from noon to midnight (For Example 5.45 is a palindrome)?

Options:

- **(A)** `43`
- **(B)** `51`
- **(C)** `57`
- **(D)** `58`

Correct Answer: (C) `57`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
57


after 12 o clock 12:21.......=1


from 1 to 9 it is 1.01, 1.11, 1.21, 1.31, 1.41, 1.51.= 6


similarly 2.02,2.12,.................


6*9=54+1=55


after 10 o clock 10:01, 11:11


Ans: 55+2=57
```

Question 8 (PrepInsta quiz-1 - Q8)

Question Stem:
One quarter of the time till now from midnight and half of the time remaining from now up to midnight adds to the present time. What is the present time?

Options:
- **(A)** `9:38`
- **(B)** `9:56`
- **(C)** `9:36`
- **(D)** `9:16`

Correct Answer: (C) `9:36`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
x is present time


1/4(x)+1/2(24-x)=x


solve it then x = 9:36
```

Question 9 (PrepInsta quiz-1 - Q9)

Question Stem:
A clock loses 1% time during first week and then gains 2% time during the next one week. if the clock was set

right at 12noon on Sunday what will be the the time exactly that the clock will show 14days from the time it was set right?

Options:

- **(A)** `1 : 30 : 48 P.M.`
- **(B)** `1 : 40 : 48 P.M.`
- **(C)** `1 : 40 : 38 P.M.`
- **(D)** `1 : 30 : 48 A.M.`

Correct Answer: (B) `1 : 40 : 48 P.M.`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
The clock loses 1% time during the first week. In a day there are 24 hours and in a week there are 7 days. Therefore, there are 7 * 24 = 168 hours in a week.


If the clock loses 1% time during the first week, then it will show a time which is 1% of 168 hours less than 12 Noon at the end of the first week = 1.68 hours less.


Subsequently, the clock gains 2% during the next week. The second week has 168 hours and the clock gains 2% time = 2% of 168 hours = 3.36 hours more than the actual time.


As it lost 1.68 hours during the first week and then gained 3.36 hours during the next week, the net result will be a -1.68 + 3.36 = 1.68 hour net gain in time.


So the clock will show a time, which is 1.68 hours more than 12 Noon two weeks from the time it was set right. 1.68 hours = 1 hour and 40.8 minutes = 1 hour + 40 minutes + 48 seconds.


i.e. 1 : 40 : 48 P.M.
```

Question 10 (PrepInsta quiz-1 - Q10)

Question Stem:

In a clock the long hand is of 8cm and the short hand is of 7cm. if the clock runs for 4 days find out the total distance covered by both the hands?

Options:

- **(A)** `1296 pie`
- **(B)** `1380 pie`
- **(C)** `84 pie`
- **(D)** `1648 pie`

Correct Answer: (D) `1648 pie`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
4 days = 96 hours


long hand makes 1 round (i.e 2*pi*r) in 1 hour, so 96 rounds in 96 hours


short hand makes 1 round in 12 hours, so 8 rounds in 96 hours


Hence total distance = (96*2*pi*8) + (8*2*pi*7) = 1648*pi
```
